S^{2}GFormer: A Transformer and Graph Convolution Combining Framework for Hyperspectral Image Classification

Shiqi Huang, Yao Ding, Zhili Zhang, Aitao Yang, Shujun Yang, *Member, IEEE*, Yaoming Cai, *Member, IEEE*, and Weiwei Cai, *Member, IEEE*

***Abstract*— Transformer-based methods have a great ability to model nonlocal interactions between spectral and spatial information, while the local features are easily ignored. Graph convolutional neural networks (GCNs) tend to do well in exploiting neighborhood vertex interactions based on their unique aggregation mechanism, while the ability to extract global information is limited. In this article, we study to comprehensively utilize the advantages of transformer and graph convolution by combining the two structures into a unified Transformer (Graphormer) to construct both local and global interactions for hyperspectral image (HSI) classification, and spatial–spectral features enhanced Graphormer framework (S^{2}GFormer) is proposed. Specifically, a follow patch mechanism is first proposed to transform the pixel in HSI to patches while preserving the local spatial features and reducing the computational cost. Moreover, a patchwise spectral embedding block is designed to extract the spectral features of the patch, in which a neighborhood convolution is inserted for comprehensive spectral information extraction. Finally, a multilayer Graphormer Encoder module is proposed to extract the representative spatial–spectral features from the patch for HSI classification. In our network, we jointly integrate the three aforementioned parts into a unified network, and each component benefits the other. The experimental results demonstrate its suitability for HSI classification when compared with other state-of-the-art (SOTA) classifiers, particularly in scenarios with very limited labeled samples. The code of S^{2}GFormer will be made publicly available at: https://github.com/DY-HYX.**

Received 13 July 2024; revised 22 August 2024 and 26 September 2024; accepted 28 October 2024. Date of publication 30 October 2024; date of current version 15 November 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 41404022 and Grant 62401632, in part by Guangdong Province Key Construction Discipline Scientific Research Capacity Improvement Project under Grant 2022ZDJS135, in part by the Innovation Team Project of General Universities in Guangdong Province under Grant 2024WCXTD011, and in part by the Research Initiation Fund for Senior Talents of Jiangsu University under Grant JDKQ20240406. *(Corresponding authors: Yao Ding; Zhili Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2024.3488202

***Index Terms*— Follow patch mechanism, Graphormer Encoder, hyperspectral image (HSI) classification, patchwise spectral embedding.**

NOMENCLATURE

$G = (V; E)$	Graph with node set and edges set.
n	Number of graph node.
$h_i^{(l)}$	Feature presentation at layer l.
x_i	Node feature vector.
Q, K, and V	Query, key, and value.
d, d_i	Dimension of hidden feature.
$(x_{\text{center}}, y_{\text{center}})$	Coordinate of a classified pixel.
(x_p, y_p)	Upper left corner coordinate of follow patch.
(x_q, y_q)	Lower right corner coordinate of follow patch.
W	Learnable weight matrix.
N	Total number of spectrums.
W, H	Width and height of the image.
X^i_{group}	Neighboring spectrum of the ith spectrum.
σ	Nonlinear activation function.
T	Number of epochs.
P_n	Number of patches.
L	Learning rate.
V_d	Number of Graphormer Encoder module depth.
V_h	Number of self-attention (SA) heads.
$\mathcal{N}(\cdot)$	Neighborhood of graph node.

I. INTRODUCTION

WITH the increasing demand for fine recognition of ground objects, many satellites with hyperspectral sensors have been launched, making it easier to obtain hyperspectral images (HSIs). Due to the HSI contains hundreds of contiguous spectral channels [1], [2], [3], which means that HSIs can identify a variety of ground objects difficult to identify in the conventional optical remote sensing image. Due to these unique characteristics, HSIs have drawn more and more attention in academic research and various applications [4], [5], e.g., geology [6], agriculture [7], forestry [8], military [9], and other fields.

HSI classification is still a challenging task with the spectral heterogeneity and the complex spatial structure of land covers.

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In early studies, many traditional methods, e.g., support vector machine (SVM) [10], k-nearest neighbor [11], Bayesian estimation method [12], random forests (RFs) [13], multinomial logistic regression [4], [14], and parse and collaborative representation [15], are first designed with handcrafted features to solve high-dimensional and nonlinear problems in HSI classification [16], [17]. These methods can well solve the classification problem with small samples. However, they often encounter performance bottlenecks for the complex training set and large-scale training size [18]. The reason is that these traditional methods are highly dependent on the prior information and robustness deficiency, leading to limited data fitting and relatively poor representation capabilities [19], [20].

Deep learning (DL) technology [e.g., recurrent neural networks (RNNs), stacked autoencoders (AEs), and convolutional neural networks (CNNs)] has the advanced image processing technology development in various fields and contributed to remote sensing image processing the technological innovation. To overcome the shortcomings of traditional methods, numerous deep-learning-based classifiers have been designed for HSI classification with their deep spatial–spectral feature extraction ability. Chen et al. [21] introduce SAEs into HSI classification by flatting the training patches into 1-D features as the input extracting the spatial–spectral hierarchical features of HSI. Hang et al. [22] and Mou et al. [23] employed the RNNs for HSI classification by modeling the pixels in HSI as sequence data. Owing to the good performances of CNNs in the image processing, a large number of CNN-based networks were developed. The first application was 1-D CNN proposed by Hu et al. [24] with five convolutional layers. However, this method ignored the spatial features of HSI. After that, to extract the spatial–spectral features, the 2-D CNN was utilized by Zhao and Du [25] and Yue et al. [26]. To convert HSI into an image that can be processed by 2-D CNN, the dimensionality reduction method must be applied first to reduce the spectral dimension of HSI, resulting in insufficient use of rich spectral features. Furthermore, Yang et al. [27] designed a dual-branch network to better extract the joint spatial–spectral information from HSI by fusing 1-D CNN and 2-D CNN structures. To better extract spatial–spectral fusion features, in [28] and [29], the 3-D CNN has been applied to HSI classification. Furthermore, Roy et al. [30] proposed a new classifier named HybridSN combining the advantages of 3-D CNN and 2-D CNN. To better exploit the performance of CNN, numerous complex structure CNN-based networks have been designed for HSIs, e.g., pyramid [31], dense [32], and residual [33] structures.

However, the aforementioned DL classifiers contain their disadvantages in HSI classification. For example, RNNs can model the pixels in HSIs as sequence data to represent bands, they are exceedingly relying on ordered spectral bands resulting unable to deal with long-term dependence. Although CNNs have greatly advanced the development of HSI processing technology and improved classification accuracy, they excessively capture local spatial features of an HSI failing to extract useful spectral-sequence features.

Compared with CNNs, graph convolutional neural network (GCN) is capable of aggregating of neighborhood node features in non-Euclidean space. Hence, the remote spatial dependency of different nodes is established, and the ability to extract spatial features has been improved [34]. Recently, applying GCN to HSI classification has received widespread attention. For example, a GCN network with spatial–spectral information named S^2GCN was proposed by Qin et al. [35]. However, the strategy of taking every pixel as a node is computationally intensive and inefficient. In [36], a new miniGCN is proposed to reduce the computation, and the large-scale graph can be trained in a small-scale manner. Moreover, Ding et al. [37] transform the HSI to a superpixel structure as a graph node, and adopted the GraphSage mechanism to extract the high-level features of HSI. In [38], a multiscale dynamic graph convolutional network was proposed, which can flexibly explore the relations among graph-style data. Yu et al. [39] proposed a novel two-branch deeper GCN (TBDGCN) to combine the advantages of superpixel-based GCN and pixel-based CNN, which can simultaneously extract superpixel- and pixel-level features of HSIs. Dong et al. [16] fully utilized the characteristics of pixel-based CNN and superpixel-based graph attention network (GAT) to extract the local and nonlocal spatial information to improve the HSI classification. Ding et al. [40] propose a cross-domain few-shot learning framework based on global–local graph attention (GLGAT-CFSL) networks, which comprehensively reduces domain shift through global-to-local domain alignment. Although GCN-based methods achieve outstanding performances in HSI classification, they still have some drawbacks. To construct the graph adjacency matrix of GCN is computationally costly and inefficient. In addition, only the spatial relationship is established with the adjacency matrix, the spectral correlation is ignored.

Recently, the excellent performance of Vision Transformer (ViT) [41] in image processing, has attracted the strong interest of scholars in the field of remote sensing. Thus, some ViT-based classifiers have been designed for HSI classification. In [42], a HSI-bidirectional encoder representations from transformers (BERT) is designed to overcome the poor generalization and weak flexibility of traditional DL methods with dynamic and flexible inputs,. He et al. [43] developed a spatial–spectral transformer (SST) network by combining VGGNet-like structure and dense transformers to model adjacent spectra's relationship. In [44], a SpectralFormer (SF) was proposed to capture the local sequence information of adjacent spectra embedded by group spectrum, and a cross-layer adaptive fusion (CAF) module was proposed to avoid the information loss during feature extraction. To capture spatial–spectral features, Sun et al. [45] designed a Gaussian weighted feature tokenizer to perform feature transformation. Xie et al. [46] proposed the S3FFT method, in which spatial attention and efficient channel attention (ECA) modules were employed for the extraction of shallow spatial–spectral features with which the high-level semantic features can be represented better. To address the problem of overdispersion of features extracted by the multihead self-attention (MHSA) in Transformer, Mei et al. [47] introduced a grouped pixel embedding (GPE) to better aware the local spatial–spectral features. Jiang et al. [48] proposed a graph generative structure-aware Transformer (GraphGST) to learn the local graph features while preserving the global features

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TABLE I
DETAILS OF EACH LAND-COVER CLASS OF THREE DATASETS USED FOR EVALUATION IN THE EXPERIMENTS

NO.	Indian Pines			Salinas			UH2013		
	Category Name	Training	Tesing	Category Name	Training	Tesing	Category Name	Training	Tesing
1	Asphalt	15	31	Weed 1	30	1979	Healthy-grass	30	1221
2	Corn-notill	30	1398	Weed 2	30	3696	Stressed-grass	30	1224
3	Corn-mintill	30	800	Fallow	30	1946	Synthetic-grass	30	667
4	Corn	30	202	Fallow-plow	30	1364	Tress	30	1214
5	Grass-pasture	30	453	Fallow-smooth	30	2648	Soil	30	1212
6	Grass-trees	30	700	Stubble	30	3929	Water	30	295
7	Grass-pasture-mowed	15	13	Celery	30	3549	Residential	30	1238
8	Hay-windrowed	30	448	Grapes untrained	30	11241	Commercial	30	1214
9	Oats	15	5	Soil	30	6137	Road	30	1222
10	Soybean-notill	30	942	Corn	30	3248	Highway	30	1197
11	Soybean-mintill	30	2425	Lettuce-4wk	30	1038	Railway	30	1205
12	Soybean-clean	30	563	Lettuce-5wk	30	1897	Parking-Lot 1	30	1203
13	Wheat	30	175	Lettuce-6wk	30	886	Parking-Lot 2	30	439
14	Woods	30	1235	Lettuce-7wk	30	1040	Tennis-Court	30	398
15	Buildings-Grass-Trees-Drives	30	356	Vineyard-untrained	30	7238	Running-Track	30	630
16	Stone-Steel-Towers	30	63	Vineyard-trellis	30	1777			

of the Transformer. Jia et al. [49] proposed a center-masked transformer (CMT) approach, in which a local-to-global token embedding (L2GTE) framework is used.

Transformers are effective in modeling spectral information of HSI. However, the following drawbacks should be concerned: first, compared with normal images, HSIs contain a multitude of spectrums. Extracting the information from rich spectral information may lead the extracted features to be too scattered, and the local adjacent spectral band contributions in HSI for classification [50] are ignored. Furthermore, the Transformer overly focuses on global information and cannot construct relationships between adjacent nodes, resulting in limited MHSA's ability to extract spectral information. In addition, the most of inputs of the spatial–spectral fusion Transformer are patches, which may lead to the misclassification of pixels across class boundaries, as the input is not a single pixel, and the high computational complexity also be accompanied by undifferentiated input neighborhood information.

Transformer-based methods have a great ability to model nonlocal interactions among spectral and spatial information, whereas GCNs tend to do well in exploiting neighborhood vertex interactions based on their unique aggregation mechanism; therefore, we are considering whether it is possible to embed GCN in the Transformer to leverage the advantages of both to improve the classification accuracy of HSI. In this article, we study comprehensively utilize the advantages of transformer and graph convolution by combining the two structures into a unified Transformer (Graphormer) to model both global and local interactions for HSI classification, and spatial–spectral features enhanced Graphormer framework (S^{2}GFormer) is proposed. Specifically, we first propose a follow patch mechanism to transform the pixel in HSI into patches while preserving the local spatial features. We should also note that the follow patch also provides a new idea for extracting the spatial structure features of image edge pixels. After that, to extract the spectral features of the patch, a patchwise spectral embedding block is designed, in which a neighborhood convolution, different from the available spectral information extraction methods, is proposed to extract the comprehensive spectral information of HSI. Finally, the multilayer Graphormer Encoder module is developed to extract the representative spatial–spectral features from the patch for HSI classification. Notably, the Graphomer Encoder is newly designed and is essentially different from Transformer. In this study, we jointly integrate the three aforementioned parts into a unified end-to-end network, and each component benefits the other.

TABLE II
PRESET OPTIMAL PARAMETERS IN S^{2}GFORMER

Dataset	T	P_n	L	V_d	V_h
IP	200	9×9	5×10^{-4}	3	4
Salinas	200	9×9	5×10^{-4}	3	4
UH 2013	200	9×9	5×10^{-4}	3	4

We can summarize the contributions as follows.

1) To comprehensively utilize the advantages of Transformer and graph convolution and extract the representative spatial–spectral features from the patches, a multilayer Graphormer Encoder module is proposed by combining the two structures into a unified module to construct both global and local interactions.
2) To mine the relationship between HSI pixels and effectively preserve the local spatial features, a follow patch mechanism is designed. Then, we can reduce the number of graph nodes and simplify the computational complexity of the proposed method.
3) To better fuse local information while adopting the Transformer to establish global attention, a patchwise spectral embedding block is developed with a neighborhood convolution.

The remainder of this article is organized as follows. Some related works and concepts concerning the proposed method are reviewed in Section II. In Section III, we formulate the theoretical basis of S^{2}GFormer, we also present our training strategy here. We conduct the accuracy experiments and uncertainty estimation in Section IV. Finally, a comprehensive conclusion is drawn, and a brief outlook is discussed in Section V.

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TABLE III
QUANTITATIVE EXPERIMENTAL CLASSIFICATION RESULTS ON IP, THE BEST AND SECOND-BEST PERFORMANCES ARE MARKED IN BOLD AND UNDERLINE

No.	2D CNN	CRNN	miniGCN	S^2GCN	GLGAT-CFSL	L2GTE	SF	SSFTT	S^2GFormer
1	87.01±0.00	75.36±0.26	68.32±5.39	76.83±3.97	**100.00±0.00**	98.65±1.35	47.31±13.52	**100.00±0.00**	**100.00±0.00**
2	41.18±6.81	38.35±5.97	48.17±7.31	57.29±6.29	78.83±1.	79.28±2.02	50.74±5.01	77.23±2.92	**85.77±1.54**
3	46.92±8.72	51.23±8.01	46.28±7.05	61.02±5.37	82.62±0.61	80.36±5.42	57.04±6.71	77.46±10.09	**95.62±0.28**
4	88.89±2.40	79.66±3.87	82.03±4.24	78.66±1.82	96.34±0.73	75.32±0.90	84.06±6.35	**98.39±1.39**	98.07±1.49
5	73.36±6.93	70.62±4.01	76.02±2.19	80.60±1.85	**96.34±0.73**	75.32±0.90	72.33±6.60	93.75±4.06	95.36±0.18
6	88.29±4.15	85.27±2.64	89.19±3.09	92.28±2.59	95.05±1.26	**98.51±0.67**	82.29±0.91	96.67±1.36	98.43±0.34
7	**100.00±0.00**	**100.00±0.00**	95.13±1.42	98.33±1.92	99.51±0.29	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
8	97.02±1.00	98.66±1.28	87.06±1.57	91.36±0.61	97.61±1.36	**99.78±0.27**	96.35±1.24	**99.78±0.32**	**99.78±0.12**
9	**100.00±0.00**	**100.00±0.00**	93.00±1.97	**100.00±0.00**	99.57±0.75	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
10	57.93±3.73	58.03±5.59	62.82±4.33	68.26±3.82	80.34±0.90	90.68±1.32	71.09±4.68	87.30±2.73	**91.08±1.71**
11	59.64±3.73	43.92±5.27	66.08±2.62	71.08±2.01	81.57±0.76	88.37±0.81	53.46±9.35	67.68±7.54	**90.93±1.37**
12	62.94±5.66	58.30±4.98	51.07±5.88	62.35±3.21	71.25±0.55	85.39±1.53	43.64±4.48	83.37±2.76	**90.94±0.52**
13	99.05±0.27	96.07±0.95	89.31±0.32	91.06±1.00	97.72±0.99	**100.00±0.00**	97.71±1.62	99.62±0.54	**100.00±0.00**
14	83.13±0.60	81.28±1.11	86.07±0.92	91.06±1.38	97.60±0.35	96.78±0.86	76.60±5.15	91.77±1.61	**99.60±0.36**
15	80.43±6.91	76.54±5.90	82.39±1.58	87.37±1.93	70.29±1.66	98.38±1.26	85.49±3.32	95.51±3.58	**99.44±0.75**
16	100.00±0.00	96.01±1.62	76.92±1.25	85.93±1.72	**100.00±0.00**	**100.00±0.00**	98.94±1.50	96.30±5.24	**100.00±0.00**
OA (%)	65.83±1.78	66.28±0.99	68.73±2.61	75.24±1.97	84.32±0.82	90.38±1.30	65.24±2.09	82.96±1.21	**93.55±0.47**
AA (%)	79.12±1.21	75.58±1.92	74.99±0.41	80.84±0.81	90.21±1.38	92.61±0.92	76.07±0.95	91.55±0.97	**96.56±0.35**
Kappa	61.38±1.99	68.21±0.48	68.36±1.94	74.71±2.38	85.49±0.82	86.19±0.260	61.06±2.10	80.71±1.29	**92.62±0.53**

II. RELATED WORKS AND CONCEPTS

A. Notations of Paper

In the Nomenclature, we demonstrate the main notations adopted in this article and corresponding descriptions.

B. Graph Convolutional Neural Network (GCN)

Given a graph $G = (V; E)$ where $V = \{v_1, v_2, \ldots, v_n\}$, the number of graph nodes is $n = |V|$. Let a node v_i with a feature vector x_i. Graph neural networks (GNNs) aim to learn the representation of nodes and graphs.

Generally, GNN can be abstracted into two steps, including aggregating and combining, which basically follow the framework of message passing. The representation of graph nodes is updated by aggregating the representation of the first or higher order neighbors of nodes. Let $h_i^{(l)}$ be the feature presentation of graph node v_i at layer l, and $h_0^{(l)} = x_i$. We can express the lth iteration of aggregation of GNN formulated by aggregate–combine as follows:

$$a_i^{(l)} = \text{AGGREGATE}^{(l)}\left(\left\{h_j^{(l-1)} : j \in \mathcal{N}(v_i)\right\}\right)$$
$$h_i^{(l)} = \text{COMBINE}^{(l)}\left(h_i^{(l-1)}, a_i^{(l)}\right) \quad (1)$$

where $\mathcal{N}(v_i)$ is the first or higher order neighbor set of v_i. In (1), we can adopt different AGGREGATE functions, e.g., MEAN, MAX, and SUM to extract the neighbor's information. In addition, the COMBINE function is used to fuse the neighbor's information gathered by the AGGREGATE function into the node representation.

Furthermore, only obtaining the hidden vector of each node is not enough to represent the entire graph G, so a graph READOUT function is needed to represent the hidden vector of graph nodes as a vector of uniform length h_G, namely,

$$h_G = \text{READOUT}\left(\left\{h_i^{(L)} \middle| v_i \in G\right\}\right). \quad (2)$$

In (1), the READOUT function can perform max pooling and average pooling on the hidden vectors of each node to obtain features. We can also introduce a god node to connect the nodes in each graph and take the hidden vector of the god node at the last layer as the representation. It can also be different hierarchical pooling, edge pooling, Set2Set, and other methods.

C. Vision Transformer

The Transformer architecture contains a sequence of Transformer layers containing an SA module. Given an SA module input $\mathbf{H} = [h_1^{\mathrm{T}}, h_1^{\mathrm{T}}, \ldots, h_n^{\mathrm{T}}]^{\mathrm{T}} \in \mathbb{R}^{n\times d}$, where d and $h_i \in \mathbb{R}^{1\times d}$ denote the hidden dimension and the hidden representation at position i, respectively. The input H are mapped into three matrices $\mathbf{Q}$(Query), $\mathbf{K}$(Key), and $\mathbf{V}$(Value) by sharing the weights $W_q \in \mathbb{R}^{d\times d_k}$, $W_k \in \mathbb{R}^{d\times d_k}$, and $W_v \in \mathbb{R}^{d\times d_v}$, respectively. We can formula the SA layer as follows:

$$\mathbf{Q} = \mathbf{H}\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{H}\mathbf{W}_K, \quad \mathbf{V} = \mathbf{H}\mathbf{W}_V \quad (3)$$
$$\mathbf{A} = \frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_K}}, \quad \text{Attn}(\mathbf{H}) = \text{softmax}(\mathbf{A})\mathbf{V} \quad (4)$$

where $\mathbf{A}$ is the matrix that captures the similarity between Q and K.

III. PROPOSED S^2GFORMER FOR HSI CLASSIFICATION

In this section, we illustrate the architecture and each step of the proposed S^2GFormer in detail. Specifically, we first introduce the overall architecture of S^2GFormer (Section III-A). Then, HSI preprocessing and follow patch mechanism are described (Section III-B). Subsequently, the patchwise spectral embedding block is introduced (Section III-C). Furthermore, the multilayer Graphormer Encoder is developed, including MHSA and graph residual block. Finally, the training and optimization details of S^2GFormer are detailed.

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TABLE IV
QUANTITATIVE EXPERIMENTAL CLASSIFICATION RESULTS ON SALINAS, THE BEST AND SECOND-BEST PERFORMANCES ARE MARKED IN BOLD AND UNDERLINE

No.	2D CNN	CRNN	miniGCN	S²GCN	GLGAT-CFSL	L2GTE	SF	SSFTT	S²GFormer
1	98.77±0.80	99.34±0.53	95.20±0.82	99.01±0.44	99.00±0.82	**100±0.00**	94.74±0.93	**100±0.00**	**100±0.00**
2	98.70±0.71	99.17±0.21	98.61±0.59	99.18±0.59	99.12±1.66	99.65±0.35	98.82±0.49	99.49±0.46	**99.86±0.13**
3	90.00±1.19	96.54±1.39	94.15±0.94	99.50±0.46	99.82±0.18	92.85±0.61	92.17±1.99	90.10±7.84	**99.98±0.03**
4	98.75±0.36	97.32±0.27	93.07±0.82	99.11±0.55	99.32±0.94	98.37±1.24	96.26±0.94	**99.76±0.30**	99.29±0.04
5	95.20±1.07	**98.75±0.89**	87.23±1.14	97.55±2.35	96.27±0.99	96.12±0.92	89.41±1.83	97.13±2.30	96.11±1.35
6	99.40±0.39	99.19±0.77	95.13±0.86	99.32±0.35	99.37±1.28	99.48±0.26	99.20±0.72	**99.71±0.31**	99.50±0.21
7	**99.38±0.37**	98.67±1.30	90.28±0.62	90.06±0.27	99.20±1.60	99.29±0.62	97.91±1.99	99.03±0.50	99.30±0.49
8	72.55±6.82	72.38±3.98	69.31±0.97	70.68±5.20	**88.79±1.52**	83.51±1.85	73.07±1.12	73.95±3.67	**92.59±1.02**
9	93.30±0.93	97.29±0.61	96.00±1.40	98.32±1.79	99.60±0.41	99.47±0.53	93.77±1.42	98.59±1.35	**99.62±0.37**
10	88.24±0.95	91.44±1.64	88.29±2.69	90.97±2.59	88.32±1.83	93.25±1.79	91.39±1.83	94.28±1.14	**95.83±0.43**
11	93.96±2.26	96.82±0.78	96.85±0.26	98.00±1.65	96.51±0.68	97.38±1.26	92.36±3.31	98.65±0.27	**99.79±0.12**
12	99.33±0.22	99.21±0.32	96.73±0.71	99.56±0.59	94.32±1.72	97.21±0.29	99.05±0.58	99.95±0.04	**99.66±0.21**
13	99.06±0.93	97.29±0.86	95.29±1.31	97.83±0.72	95.05±0.85	92.30±1.80	**99.40±0.43**	97.86±1.28	99.32±0.65
14	96.73±0.95	95.10±1.73	93.66±1.27	95.75±1.65	93.06±0.27	96.07±0.92	96.60±0.95	98.91±0.16	**99.42±0.63**
15	71.35±3.32	76.33±4.62	71.49±2.61	70.36±3.62	87.18±1.87	80.13±2.60	81.58±2.20	81.10±3.63	**93.27±0.16**
16	92.61±1.97	97.99±0.61	95.08±1.25	96.90±1.97	91.33±0.92	97.41±1.88	93.72±2.04	96.96±2.05	**98.26±0.60**
OA (%)	87.59±1.49	87.64±0.83	86.75±1.97	88.39±1.01	91.82±1.33	92.26±0.73	88.80±0.21	90.67±1.52	**96.92±0.36**
AA (%)	92.96±0.56	94.55±0.57	91.02±0.82	94.30±0.47	95.37±1.24	95.59±0.87	93.09±0.19	95.34±1.11	**98.23±0.07**
Kappa	86.21±1.64	86.72±1.01	83.29±1.48	87.10±1.12	92.76±1.62	95.31±0.46	87.57±0.24	89.64±1.68	**95.62±0.31**

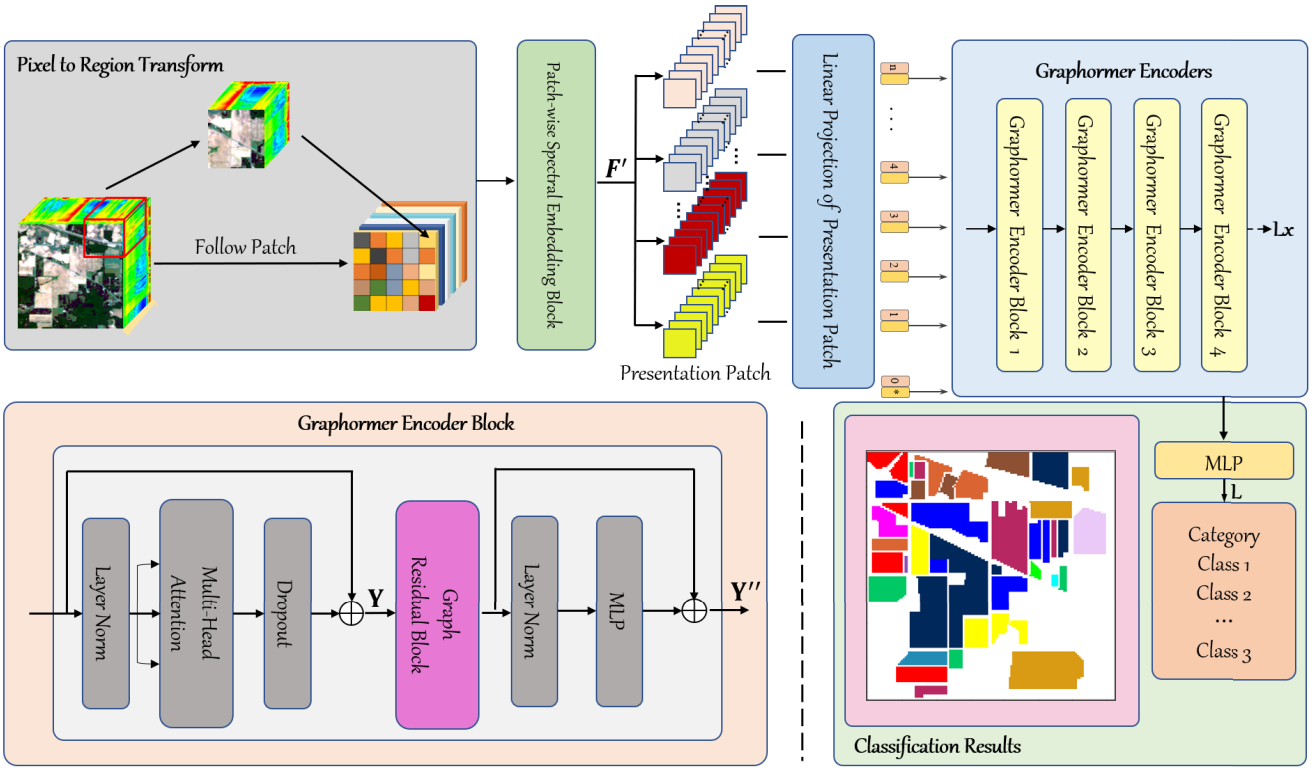


Fig. 1. Illustration of S²GFormer for HSI classification. S²GFormer mainly consists of three parts, i.e., pixel-to-region transform module, patchwise spectral embedding block, and multilayer Graphormer Encoder module. The pixel-to-region transforms module is proposed to transform the pixels in HSI to locality-preserving regions using follow patch mechanism. The patchwise spectral embedding block is developed to extract the spectral features of the patch. The multilayer Graphormer Encoder module is designed to extract the representative spatial–spectral features from the patch for HSI classification, which is newly designed and essentially different from the Transformer.

A. Overall Architecture Description

The proposed S²GFormer mainly consists of three components, i.e., pixel-to-region transform module, patchwise spectral embedding block, and multilayer Graphormer Encoder module, and the overall framework of S²GFormer is shown in Fig. 1. In the following, the three portions will be elaborated in detail.

HSI Preprocessing and Follow Patch: The HSIs are transformed from pixels to regions while preserving the local spatial–spectral structure and reducing the computational complexity with follow patch mechanism.

Patchwise Spectral Embedding Block: The local patches are treated as the input. To adaptively extract the representative spectral features for patches while reducing the computational

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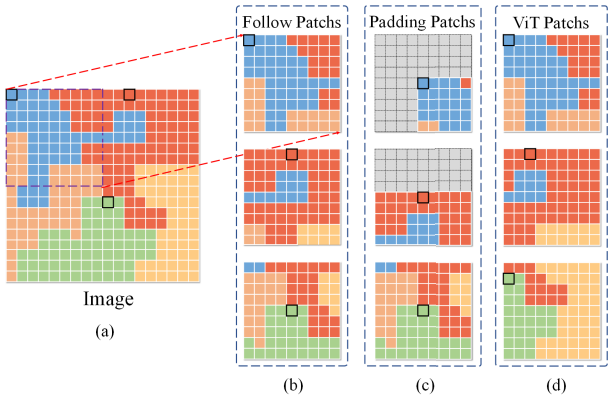


Fig. 2. Illustration of the follow patch mechanism. (a) Input image and (b)–(d) results of the follow patch method, padding patch method, and ViT patch method. Notably, the black-bordered pixels in the figure are the target pixels.

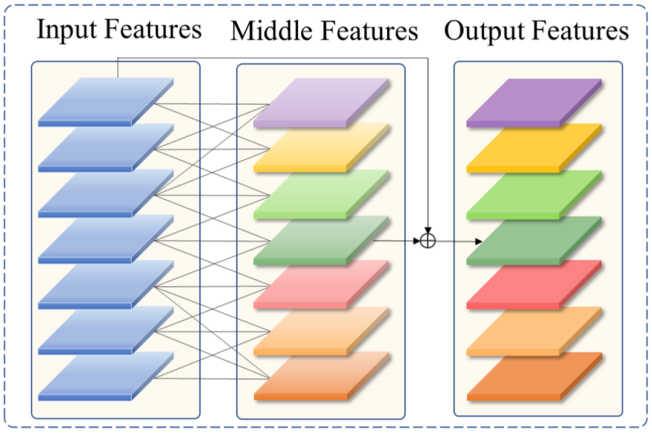


Fig. 3. Illustration of neighborhood convolution.

complexity, this module builds a neighborhood convolution mechanism by aggregating the local spectral information of each band in HSI.

Multilayer Graphormer Encoder: The spatial–spectral features are extracted for HSI classification. Furthermore, in Graphormer Encoder, both global and local spatial–spectral interactions are modeled by combining SAs and graph convolutions into a Transformer, newly designed and essentially different from the Transformer.

In our S^{2}GFormer, the HSI preprocessing and follow patch, patch block, and multilayer Graphormer Encoder are joined in an end-to-end network for HSI classification, and we interact each portion together in a unified network.

B. HSI Preprocessing and Follow Patch

It is computationally intensive if we take each pixel as input for the S^{2}GFormer with numerous pixels contained in HSI. To solve this problem, there are two main existing methods, i.e., the superpixel segmentation method [51] and the small sample method [36]. The superpixel segmentation method can segment the whole image into some superpixels, leading to reduce the number of input nodes. However, the superpixel segmentation method inputs all samples once time, which places a great burden on memory. The small sample method is inefficient and cumbersome. Furthermore, the widely used patch methods cannot fully retain the local features of HSI, which reduces the integrity of feature input. To overcome the aforementioned deficiencies, we proposed a follow patch mechanism, with which the input computational complexity of S^{2}GFormer can be reduced while preserving the local features of HSIs. Fig. 2 shows the principal comparison of the three patch methods, which demonstrates that our follow patch mechanism can make better utilize neighbor pixel information with the same patch size.

Given a classified pixel $x(x_{\text{center}},\ y_{\text{center}})$, we take the patch with the size m (m around x as an input unit into the proposed S^{2}GFormer. For a square area, we can determine the patch position by calibrating the bottom right coordinate and top left coordinate of the patch. Namely, the upper left corner coordinate of follow patch $(x_p,\ y_p)$ can be expressed as follows:

$$x_p = \begin{cases} x_c - \dfrac{m-1}{2}, & x_c > \dfrac{m-1}{2} \text{ and } x_c < W - \dfrac{m-1}{2} \\ 0, & x_c \le \dfrac{m-1}{2} \\ W - m - 1, & x_c \ge W - \dfrac{m-1}{2} \end{cases} \tag{5}$$

$$y_p = \begin{cases} y_c - \dfrac{m-1}{2}, & x_c > \dfrac{m-1}{2} \text{ and } y_c < H - \dfrac{m-1}{2} \\ 0, & y_c \le \dfrac{m-1}{2} \\ H - m - 1, & y_c \ge H - \dfrac{m-1}{2} \end{cases} \tag{6}$$

where H and W are the height and width of an image, respectively. m is an odd number. Similarly, the lower right corner coordinate $(x_q,\ y_q)$ is formed as follows:

$$x_q = \begin{cases} x_c + \dfrac{m-1}{2}, & x_c > \dfrac{m-1}{2} \text{ and } x_c < W - \dfrac{m-1}{2} \\ m - 1, & x_c \le \dfrac{m-1}{2} \\ W - 1, & x_c \ge W - \dfrac{m-1}{2} \end{cases} \tag{7}$$

$$y_q = \begin{cases} y_c + \dfrac{m-1}{2}, & y_c > \dfrac{m-1}{2} \text{ and } y_c < H - \dfrac{m-1}{2} \\ m - 1, & y_c \le \dfrac{m-1}{2} \\ H - 1, & y_c \ge H - \dfrac{m-1}{2}. \end{cases} \tag{8}$$

With $((x_p,\ y_p), (x_q,\ y_q))$, we can accurately define a patch. From the above analysis, we can observe that the proposed mechanism extends the spatial information of the image edge points and preserves richer spatial features, which is different from CNN selecting the convolutional region in the form of padding.

C. Patchwise Spectral Embedding

We proposed a follow patch mechanism to preserve the local features of the HSI with patches. We still have not solved

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TABLE V
QUANTITATIVE EXPERIMENTAL CLASSIFICATION RESULTS ON UH2013, THE BEST AND SECOND-BEST PERFORMANCES ARE MARKED IN BOLD AND UNDERLINE

No.	2D CNN	CRNN	miniGCN	S²GCN	GLGAT-CFSL	L2GTE	SF	SSFTT	S²GFormer
1	85.53±3.42	82.45±2.19	85.16±3.72	96.30±3.07	93.81±1.26	82.37±0.91	86.66±2.51	94.31±1.39	**98.24±0.62**
2	81.02±4.09	84.12±3.64	75.16±4.26	**98.57±1.47**	90.27±0.97	84.12±1.82	91.34±1.32	87.32±2.61	97.67±0.62
3	89.27±1.98	91.56±1.07	85.72±1.35	98.88±0.43	94.53±0.82	**100.00±0.00**	93.76±0.79	92.36±1.27	**100.00±0.00**
4	86.31±2.92	91.29±3.78	89.32±2.92	97.68±2.89	95.32±1.31	99.26±0.74	90.31±0.39	88.42±0.92	**100.00±0.00**
5	94.72±1.51	98.81±0.62	95.37±1.28	97.66±1.12	97.15±0.56	**100.00±0.00**	96.29±1.76	95.92±1.22	**100.00±0.00**
6	92.31±1.25	94.83±2.19	89.46±1.60	**96.84±1.17**	95.01±0.39	92.15±0.64	83.12±4.97	81.39±2.87	94.26±1.18
7	87.72±1.39	86.42±3.42	88.17±4.32	83.48±5.89	82.26±2.64	80.61±1.93	76.48±5.33	85.44±3.28	**93.21±0.97**
8	57.02±5.89	53.05±8.71	60.26±3.37	76.15±4.37	74.80±2.19	69.02±2.48	82.91±1.39	77.02±4.91	**89.52±2.82**
9	80.30±3.76	84.04±4.63	86.32±2.67	82.17±1.78	83.71±1.37	78.30±1.31	75.92±3.44	79.39±2.76	**90.16±0.76**
10	51.79±6.18	45.14±8.77	50.29±7.32	86.85±8.32	86.22±2.40	95.08±0.69	87.22±2.97	90.31±4.72	**98.67±0.88**
11	62.36±3.27	61.85±9.39	66.50±6.30	88.57±5.06	79.69±1.83	87.93±1.95	81.32±1.52	89.32±2.60	**94.02±0.82**
12	80.15±2.32	84.40±2.93	80.02±3.10	78.64±4.79	82.30±1.32	66.45±2.74	76.26±3.86	83.77±2.93	**93.62±1.25**
13	87.29±1.92	84.14±3.12	80.39±2.66	75.62±6.93	80.29±1.56	82.72±2.84	77.39±4.24	83.81±4.86	**89.92±1.02**
14	94.03±1.56	96.03±0.73	92.30±1.92	99.45±0.44	99.08±0.92	**100.00±0.00**	91.62±1.40	90.35±0.63	**100.00±0.00**
15	88.32±1.78	93.45±2.41	94.07±0.91	**98.03±1.07**	95.79±0.27	95.16±1.77	90.31±2.35	95.11±1.66	96.26±0.84
OA (%)	80.32±2.09	82.10±1.21	80.67±1.32	89.31±1.00	89.92±0.76	88.23±1.06	84.77±1.42	88.06±0.82	**94.12±0.87**
AA (%)	81.59±2.31	79.21±1.02	81.23±1.76	90.33±1.06	88.68±1.26	87.54±1.66	85.39±1.68	87.61±1.28	**95.07±0.55**
Kappa	79.37±1.85	77.61±1.19	79.32±1.53	88.44±1.08	87.80±1.37	87.19±1.42	84.06±1.83	87.26±0.93	**94.66±0.56**

the problem of extracting representative spectral features for the patch. The most mainstream method is to average the spectral values of all pixels in the patch [37]; nevertheless, this method cannot solve the problem of spectral feature information redundancy and is vulnerable to spectral noise interference. At present, the spectral feature extraction pays aggressive attention to global spectral features, while ignoring local spectral features [52]. The Transformer can establish the nonlocal dependence of spectral sequence, but local information are ignored. To better fuse local information while adopting Transformer to establish global attention, a neighborhood convolution is developed, as shown in Fig. 3.

First, the current features and adjacent spectral bands are fused, we can form the neighboring bands of the ith spectrum X_i as follows:

$$X^i_{\text{group}} = \text{Concat}(X_m, X_{m+1}, \ldots, X_i, \ldots, X_{m+k-1}) \quad (9)$$

where k is the number of neighboring spectrums, is the neighboring spectrum of the ith spectrum and N is the total number of spectrums. m is expressed as follows:

$$\begin{cases} m = i - \dfrac{k-1}{2}, & i > \dfrac{k-1}{2} \text{ and } i < N - \dfrac{k-1}{2} \\ m = 0, & i \le \dfrac{k-1}{2} \\ m = N - k - 1, & i \ge N - \dfrac{k-1}{2}. \end{cases} \quad (10)$$

After that, the original spectral information is combined with a residual cascade mechanism, which is defined as follows:

$$X'_i = W_i X^i_{\text{group}} + b_i, \quad i = 1, 2, \ldots, N \quad (11)$$

$$X''_i = X'_i + X_i \quad (12)$$

$$F = \text{Concat}\left(X''_1, X''_2, \ldots, X''_N\right) \quad (13)$$

where W_i and b_i are the weight and the bias of the ith neighborhood convolution, we can obtain the output feature X'' after residual concatenation, the output feature map F is calculated with performing a cascade operation on X''_i. To prevent overfitting, the batch normalization (BN) and rectified linear unit (ReLU) are utilized to regularize output, namely,

$$F' = \text{ReLU}(\text{BN}(F)) = \text{ReLU}\left(\gamma \frac{F - \mu}{\sigma} + \beta\right) \quad (14)$$

where F'' is the output. σ and μ are the variance and expectation of a batch, respectively. γ and β trainable parameters by network training.

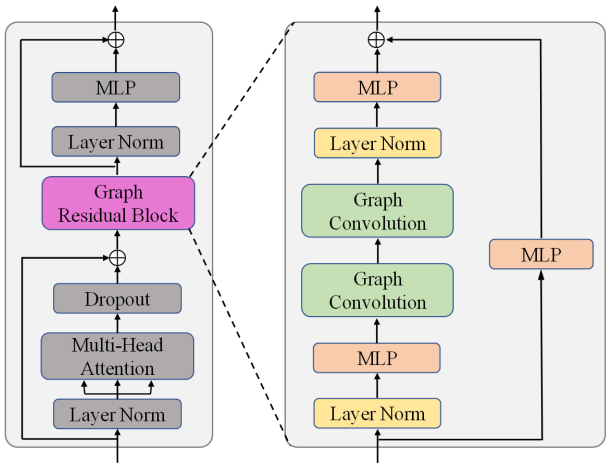


Fig. 4. Illustration of Graphormer Encoder.

D. Multilayer Graphormer Encoder

In the S²GFormer framework, we proposed a Graphormer, combining the advantages of Transformer and GCN, to extract the spectral–spatial features of presentation patches. The architecture of our proposed Graphormer Encoder is demonstrated in Fig. 4. The multilayer Graphormer Encoder is comprised of four identical blocks. Each Graphormer Encoder contains

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TABLE VI

ABLATION STUDY OF ENCODER IN GRAPHORMER WITH GRAPH RES. BLOCK ON IP, SALIANS, AND UH2013 DATASET. ✓: ADD RES. BLOCK INTO THE SPECIFIED ENCODER. ✗: NO RES. BLOCK IN THE SPECIFIED ENCODER

Encoder1	Encoder2	Encoder3	IP OA(%)	Salinas OA(%)	UH2013 OA(%)
✗	✗	✗	92.18	93.81	92.10
✓	✗	✗	92.67	94.12	92.69
✗	✓	✗	93.19	94.35	93.28
✗	✗	✓	92.86	94.97	93.95
✓	✓	✓	93.55	95.62	94.66

TABLE VII

COMPARISON OF RUNNING EFFICIENCY FOR DIFFERENT METHODS

Method	Running Time						Complexity (FLOPs)
	IP		SA		UH2013		
SF	470.37	46.05	510.01	254.53	698.79	1585.54	1.50 G
SSFTT	47.87	2.02	52.67	10.43	34.35	27.21	2.59 G
S^2 former	1.36	1.37	1.32	7.85	0.50	2.01	1.09 G

six sub-modules, i.e., a multilayer perceptron (MLP), a layer norm, a graph residual block, a layer dropout, an MHSA module, and a second layer norm. In our Graphormer Encoder, a graph convolution is embedded into the traditional Transformer encoder to model fine-grained local interactions, P which is different from the traditional Transformer $\mathbf{P}$ encoder.

1) Multihead Self-Attention: In our Graphormer Encoder, an MHSA module is employed to learn contextual representation with several SA functions. Given an input patch sequence $\mathbf{P} = \{F_1', F_2', F_3', \ldots, F_n'\} \in \mathbb{R}^{n\times d}$, where d is the spectral dimension of patches. The input patch sequence is first projected to $\mathbf{Q}, \mathbf{K}$, and $\mathbf{V}$ by learning the trainable parameters $\{\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V\} \in \mathbb{R}^{d\times d}$, namely,

$$\mathbf{Q}, \mathbf{K}, \mathbf{V} = \mathbf{P}_Q, \mathbf{P}_K, \mathbf{P}_V \in \mathbb{R}^{n\times d}. \tag{15}$$

We can split three feature representations into h different subspaces, i.e., $\mathbf{Q} = [Q^1, Q^2, \ldots, Q^h]$, $Q^i \in \mathbb{R}^{n\times d}$, with which the SA can be performed for each subspace individually. Then, the subspace output $\mathbf{Y}^h = \{y_1^h, y_2^h, \ldots, y_n^h\}$ can be calculated as follows:

$$\mathbf{y}_i^h = \mathrm{Att}\left(q_i^h, K^h\right)\mathbf{V}^h \in \mathbb{R}^{\frac{d}{h}} \tag{16}$$

where Att() is the attention function, with which the semantically relevant q_i^h–K^h is that quantified by scaling softmax and dot-product. Following that, we concatenate the output $\mathbf{Y}^h \in \mathbb{R}^{n\times(d/h)}$ to form the final output $\mathbf{Y}$.

2) Graph Residual Block: MHSA is useful to extract long-range dependencies; however, the local information in complex patch sequences is easily ignored. Therefore, we propose a graph residual block to refine the local spatial–spectral features contained in the patch sequence by extracting the relationships between different patches.

In (4), the contextualized features $\mathbf{Y}$ generated by MHSA module, the local interactions can be improved by using a graph convolution, we can compute the output $\mathbf{Y}'$ as follows:

$$\mathbf{Y}' = \mathrm{GraphConv}\left(\overline{\mathbf{A}}, \mathbf{Y}; \mathbf{W}_G\right) = \sigma\left(\overline{\mathbf{A}}\mathbf{Y}\mathbf{W}_G\right) \tag{17}$$

where $\overline{\mathbf{A}} \in \mathbb{R}^{n\times n}$, W_G, and $\sigma(\cdot)$ are adjacency matrix of a graph, trainable parameters, and activation function, respectively. In our block, we adopt Gaussian error linear unit (GeLU) to perform the network nonlinearity according to BERT [53].

Following that, the output of Graphormer Encoder $\mathbf{Y}''$ can be expressed as follows:

$$\mathbf{Y}'' = \mathrm{MLP}\left(\sigma\left(\overline{\mathbf{A}}\mathbf{Y}'\mathbf{W}_G\right)\right) + \mathrm{MLP}(\mathbf{Y}). \tag{18}$$

The graph residual block can explicitly encode the graph structure in the network, thus improving the spatial–spectral locality of the feature.

E. Training and Optimization Details

HSI preprocessing is demonstrated in (5)–(8), the output is the input of the subsequent networks. Subsequently, a patchwise spectral embedding module is introduced to extract the spectral feature for patches. Finally, a multilayer Graphormer Encoder is designed to extract spatial–spectral features of patch sequence, and the output of S^{2}GFormer is. In our S^{2}GFormer, a cross-entropy error is utilized to penalize the difference between the label examples and the proposed multilayer Graphormer Encoder, which is written as follows:

$$\mathbf{L} = -\sum_{z\in y_G}\sum_{f=1}^{c}\mathbf{Y}_{zf}\ln\mathbf{O}_{Gzf}^{(\mathrm{final})} \tag{19}$$

where y_G, Y_{zf}, and C are the labeled example set, label matrix, and number of classes. The parameters are updated by using Adam gradient descent similar to [54].

F. Computational Complexity Analysis

The computational complexity of our method mainly consists of multilayer Graphormer Encoder, i.e., MHSA and graph residual block. Denote P_n being the number of patches, d being the spectral dimension of the input patch, d_i being the spectral dimension of ith layer patch, L being the number of Graphormer Encoder layers, and T being the number of epochs. The computational complexity of MHSA is $O[{P_n}^2 d_i]$. For graph residual block, the computational complexity is $O[2P_n{d_i}^2 + P_n]$. The computational complexity of single layer Graphormer Encoder is $O[P^2 d_i + 2P_n{d_i}^2 + P_n]$. Therefore, the computational complexity of each iteration is $O[\sum_{i=0}^{i=l}({P_n}^2 d_i + 2P_n{d_i}^2 + P_n)]$. If our model iterates T times, we express the overall computational complexity of S^{2}GFormer is $O[\sum_{i=0}^{i=l}({P_n}^2 d_i + 2P_n{d_i}^2 + P_n)T]$. From the expression formula of computational complexity, we can conclude that the computational complexity can be greatly reduced by reducing the number of patches.

IV. EXPERIMENTS

In this section, extensive experiments and analyses are introduced to assess the HSI classification performance of the proposed S^{2}GFormer.

A. Benchmark Datasets and Setups

In our experiments, we select three mainstream HSI datasets, including the Indian Pines (IP) dataset, the Salinas dataset, and the Houston 2013 (UH2013) hyperspectral remote sensing images, to verify the advantages of the model we proposed. We show the true-color map, ground-truth map, and reference color of three datasets in Fig. 5. The details of three

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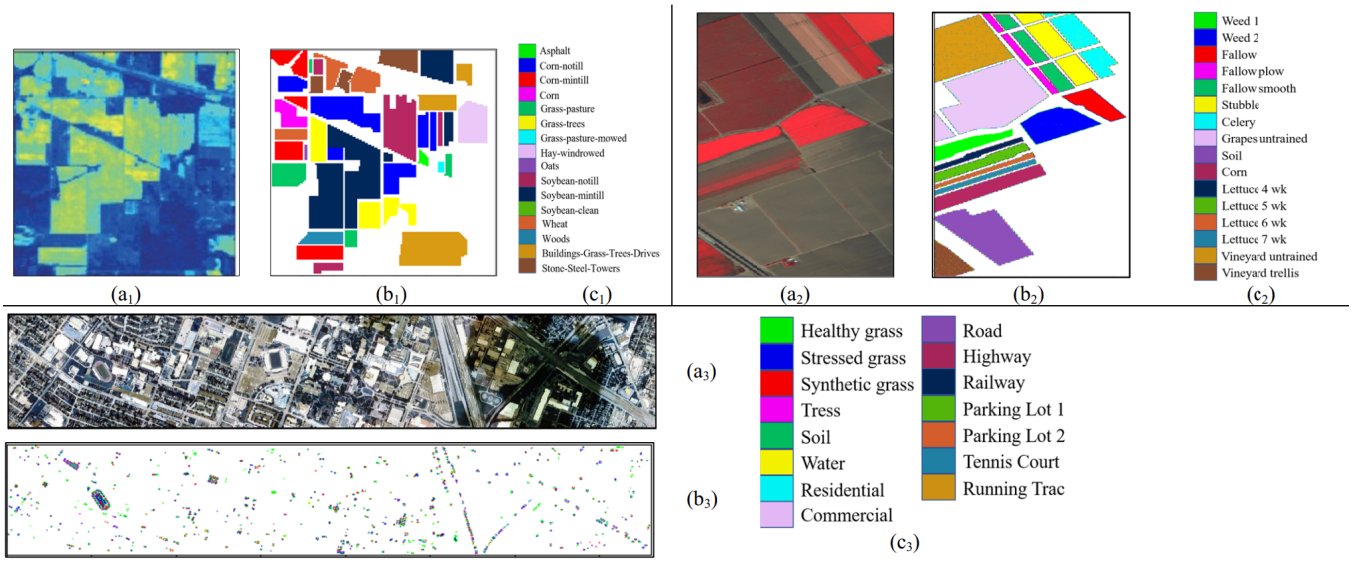


Fig. 5. Three dataset details. (a1)–(a3) False-color maps of IP, Salinas, and UH2013 datasets. (b1)–(b3) Ground-truth maps. (c1)–(c3) Corresponding reference colors.

datasets of each class used in the evaluation experiment are demonstrated in Table I.

We adopted four commonly used indices, including per-class accuracy, overall accuracy (OA), average accuracy (AA), and kappa coefficient kappa. In addition, we visualize the HSI classification maps obtained from all investigated models for qualitative comparison.

We select several recent representative baselines for the comparison experiments, including 2-D CNN [29], convolutional recurrent neural network (CRNN) [23], miniGCN [36], spectral–spatial graph convolutional network (S²GCN) [35], GLGAT-CFSL network [40], L2GTE [49], SF [44], and spectral–spatial feature tokenization transformer (SSFTT) [45]. In these methods, 2-D CNN and CRNN are CNN-based methods; miniGCN, S²GCN, and GLGAT-CFSL are GCN-based methods; L2GTE, SF, and SSFTT are state-of-the-art (SOTA) Transformer-based methods designed for HSI classification. In our S²GFormer, five main parameters, i.e., epoch number T, patch size P_n, learning rate L, depth number V_d, and heads number V_h should be preset. We demonstrate the optimal values in Table II, and how to select will be analyzed in Section IV-D. Furthermore, all investigated experiments are executed ten times for the sake of eliminating any bias caused by the random selection of training samples.

B. Comparison With SOTA Methods

In this section, following the experimental setups in Section IV-A, we perform HSI classification on three datasets, and the quantitative results are shown in Tables III–V. Furthermore, the corresponding visual comparison of the classification results generated by different classifiers are shown in Figs. 6–8.

1) Classification Results on IP Dataset: The quantitative and corresponding qualitative classification results are demonstrated in Table III and Fig. 6.

In Table III, due to extreme sample imbalance between different classes and severe noise interference, we can note that all other considered models perform poorly on IP image; however, the proposed S²GFormer achieves 93.55%, 96.56%, and 92.62% in terms of OA, AA, and Kappa, which produces an improvement of 3.17%, 3.95%, and 6.43% compared with the second-best performances achieved by L2GTE. Moreover, SSFTT and S²GCN achieve relatively good classification results, which shows that the two methods have good adaptabilities to noise interference and class imbalance sample classification. However, miniGCN and SF have no advantages over other classification methods, owing to that the two algorithms are not specifically designed for HSI classification. Due to lacking the ability to extract high-level features of HSI, CNN-based methods (i.e., 2-D CNN and CRNN) undoubtedly achieve poor results. Specifically, the OAs are 65.83% and 66.28%, respectively. In addition, although the miniGCN and S²GCN can dynamically update the graphs and fuse multiscale information for HSI classification, they cannot fully explore the relationship between spectral bands, resulting in unsatisfactory classification performance. Furthermore, compared with SSFTT, a patchwise spectral embedding block is proposed in S²GFormer to extract the presentive spectral features for HSI classification, enhancing the anti-interference of the algorithm. Hence, S²GFormer achieves the smallest among all methods in the OA, AA, and Kappa values, indicating the best stability. In terms of the qualitative visual classification maps in Fig. 6, S²GFormer obtains smoother and fewer misclassification maps compared with other competitors. We can note that SSFTT obtains relatively good classification results; nevertheless, it contains many misclassification pixels in Corn-notill, Corn-mintill, and Soybean-mintill. The above experimental results evaluate that the effectiveness of our method is better than that of competitors in the IP scene.

2) Classification Results on Salinas Dataset: The quantitative and corresponding qualitative classification results are summarized in Table IV and Fig. 7.

In Table IV, due to the Salinas dataset contains less interference noise, we can conclude that the classification results achieved by investigated methods on the Salinas

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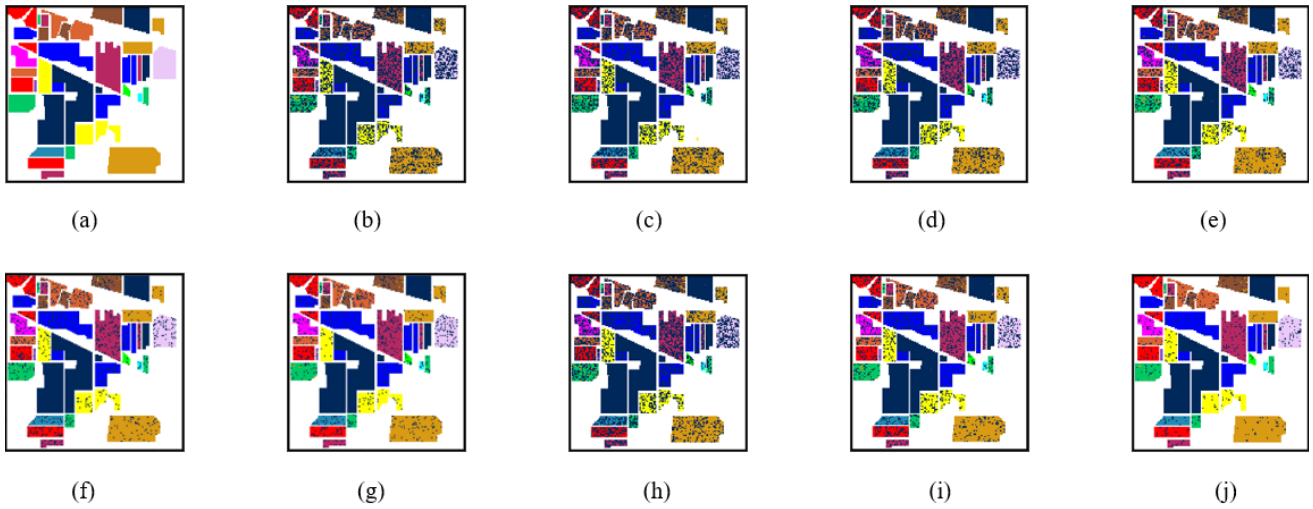


Fig. 6. Classification maps of comparison methods on IP dataset. (a) Ground-truth map. (b) Two-dimensional CNN (OA = 65.83%). (c) CRNN (OA = 66.28%). (d) miniGCN (OA = 68.73%). (e) S^2GCN (OA = 75.24%). (f) GLGAT-CFSL (OA = 84.92%). (g) L2GTE (OA = 90.38%). (h) SF (OA = 65.24%). (i) SSFTT (OA = 82.96%). (j) S^2GFormer (OA = 93.55%).

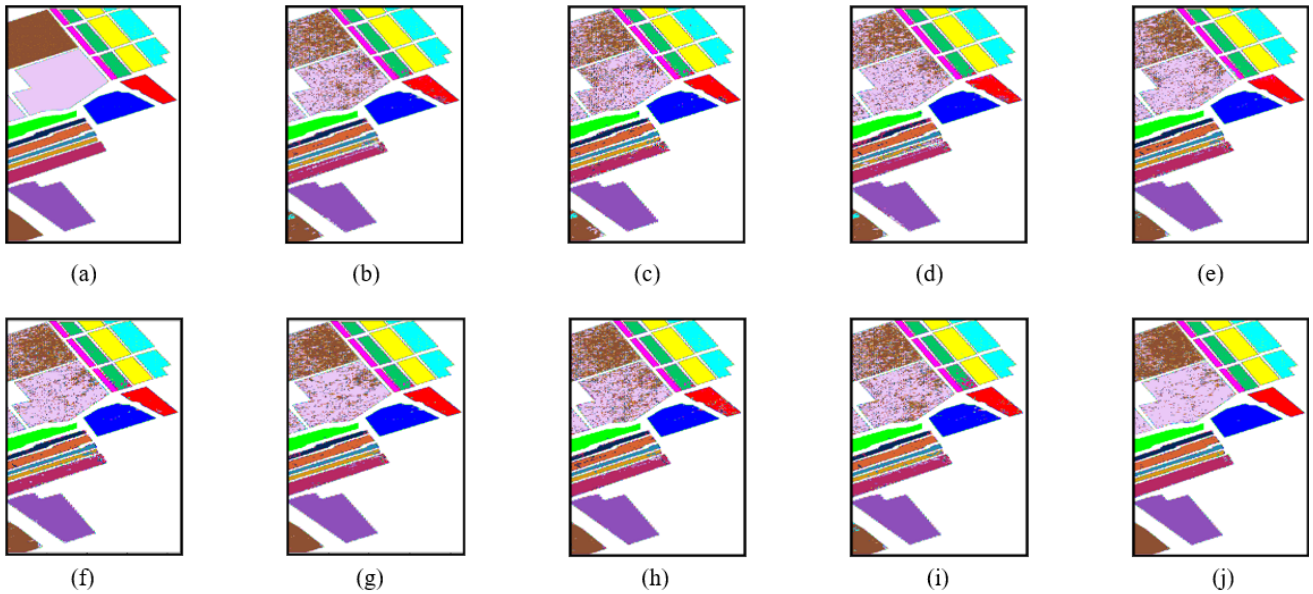


Fig. 7. Classification maps of comparison methods on Salinas dataset. (a) Ground-truth map. (b) Two-dimensional CNN (OA = 87.59%). (c) CRNN (OA = 87.64%). (d) miniGCN (OA = 86.75%). (e) S^2GCN (OA = 88.39%). (f) GLGAT-CFSL (OA = 91.82%). (g) L2GTE (OA = 92.26%). (h) SF (OA = 88.80%). (i) SSFTT (OA = 90.67%). (j) S^2GFormer (OA = 96.92%).

dataset have been greatly improved compared with the results on the IP dataset. Due to the training sample deficiency, the CNN-based classification methods achieve unsatisfactory results. For example, the OA, AA, and kappa of CRNN are 87.64%, 94.55%, and 86.72%, respectively. Surprisingly, the performance of the GCN-based classifiers for the Salinas dataset has not been significantly improved, indicating that the miniGCN and S^2GCN are unable to deal with "different objects with the same spectrum" very well. Specifically, the classification accuracy of S^2GCN in C 8 and C15 are 70.68% and 70.36%. We can well refine the nonlocal spatial–spectral features of HSIs with SA mechanism. The OA, AA, and Kappa, achieved by L2GTE are 92.26%, 95.59%, and 95.31%, respectively. The performances of our S^2GFormer are 96.92%, 98.23%, and 95.62%, which are improved by 4.66%, 2.64%, and 0.31% compared with these of L2GTE. This is mainly because our S^2GFormer method comprehensively utilizes the advantages of GCN-based methods and Transformer-based methods, thus, local spatial–spectral features are maintained while extracting nonlocal features. As the visual classification maps in Fig. 7, the visual classification map of our proposed method is closer to the ground-truth map and reduces the misclassification rate significantly.

3) Classification Results on UH2013 Dataset: The quantitative and corresponding qualitative classification results are summarized in Table V and Fig. 8, respectively.

From the quantitative results demonstrated in Table V, similar to the IP and Salinas datasets, the performance of S^2GFormer is superior to that of competitive classifiers among all key indicators, showing the good classification ability of S^2GFormer for large-scale datasets. Specifically, the results are improved by 4.81%, 4.74%, and 6.22% compared with the second-best method. Moreover, compared with miniGCN, S^2GCN and GLGAT-CFSL, Transformer-based classifiers, i.e.,

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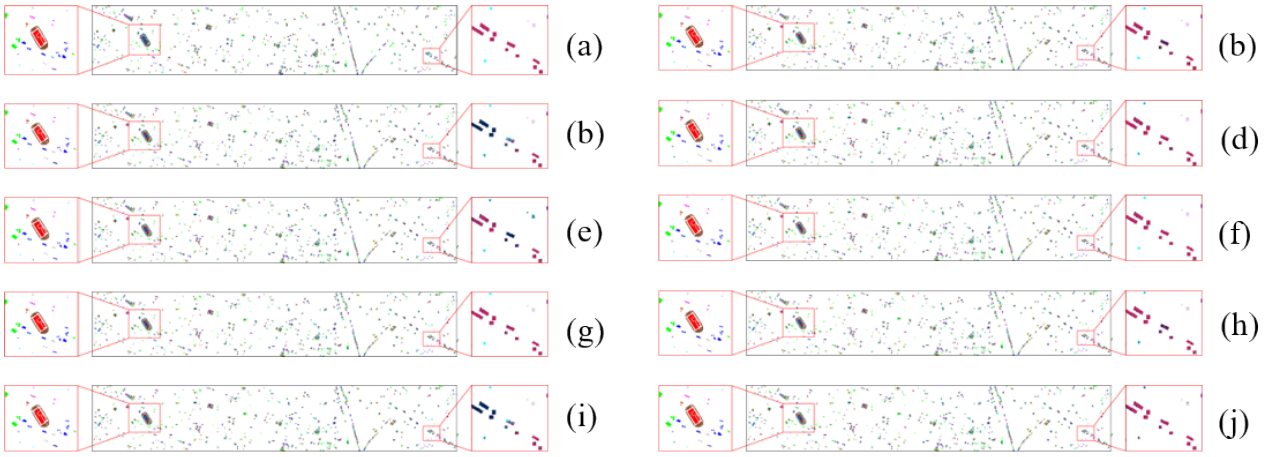


Fig. 8. Classification maps of comparison methods on UH2013 dataset. (a) Ground-truth map. (b) Two-dimensional CNN (OA = 80.32%). (c) CRNN (OA = 82.10%). (d) miniGCN (OA = 80.67%). (e) S²GCN (OA = 89.31%). (f) GLGAT-CFSL (OA = 89.92%). (g) L2GTE (OA = 88.23%). (h) SF (OA = 84.77%). (i) SSFTT (OA = 88.06%). (j) S²GFormer (OA = 94.12%).

L2GET, SF, and SSFTT, do not obtain satisfactory performances. Specifically, the OAs of the L2GET, SF, and SSFTT are only 88.23%, 84.77%, and 88.06%, respectively. Furthermore, we note that S²GCN abstains the second-best performance in all investigated classifiers, which indicates that GCN contains a good advantage for HSI classification. Furthermore, the Transformer-based classifiers (i.e., 2-D CNN and CRNN) unsatisfactory performances compared with GCN-based methods for HSI classification. As for the visual performance comparison of all investigated classifiers on the UH2013 is presented in Fig. 8. Our proposed S²GFormer obtains the best visual classification performance without doubt, verifying the advance of our S²GFormer.

C. Classification Visualization

In this section, the classification results are visualized in Fig. 9 by embedding t-distributed stochastic neighbor embedding (t-SNE) [45] into S²GFormer and SSFTT to enhance the interpretability. From the visualization, we can get the following conclusions: 1) the variance between the class center and the classification target is reduced compared with the original HSI distribution; 2) misclassifications are greatly reduced by S²GFormer; and 3) the nodes of the same class are more clustered, and the centers of different classes are farther away in the classification maps of S²GFormer. In short, the feature representation extracted by S²GFormer is interclass distance and intraclass compression compared with the competitive classifiers.

D. Influence of Label Ratio

To further evaluate the classification ability of the proposed method with a limited training sample, we randomly select different labeled samples for investigated classifiers training from 0.25% to 3% on three datasets. The other experimental setups are shown in Section IV-D. The OA index is adopted to report the performances, and the results are shown in Fig. 10. All classifiers' performances are improved with training samples increasing. Most significantly, the classification results of our S²GFormer are the best, no matter the training sample changes. Meanwhile, we can also observe

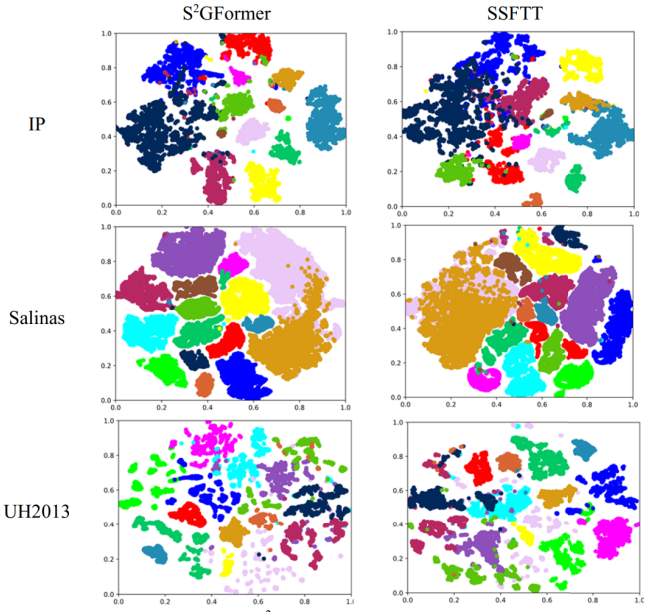


Fig. 9. Feature presentation for S²GFormer and SSFTT on three datasets.

that S²GFormer has the best adaptability to small samples, and the volatility of the results is the smallest, demonstrating the progressiveness of the classifier design. Moreover, GCN-based classifiers can better explore and exploit the relationship among the whole pixels, and Transformer-based methods can refine the global features of HSI with an SA mechanism, the two kinds of methods perform well with small samples, while the effect is not obvious as the number of training samples increasing. This result shows that only extracting local or global spectral–spatial features cannot effectively improve the performance and adaptability of the classifier confronting complex features of HSI.

E. Hyperparameter Analysis

In our S²GFormer, we should preset five main hyperparameters (i.e., T, P_n, L, V_d, and V_h), and their optimization values are shown in Table II. In this section, we explain the selection process of each hyperparameter. To tune the optimal weights,

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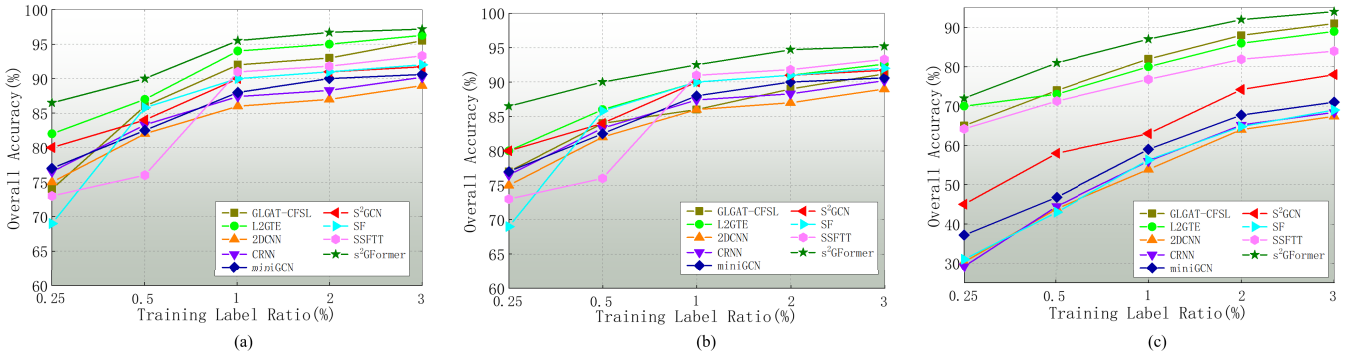


Fig. 10. Overall accuracies (%) of the investigated classifiers with different training label ratios on three datasets. (a) IP. (b) Salinas. (c) UH2013.

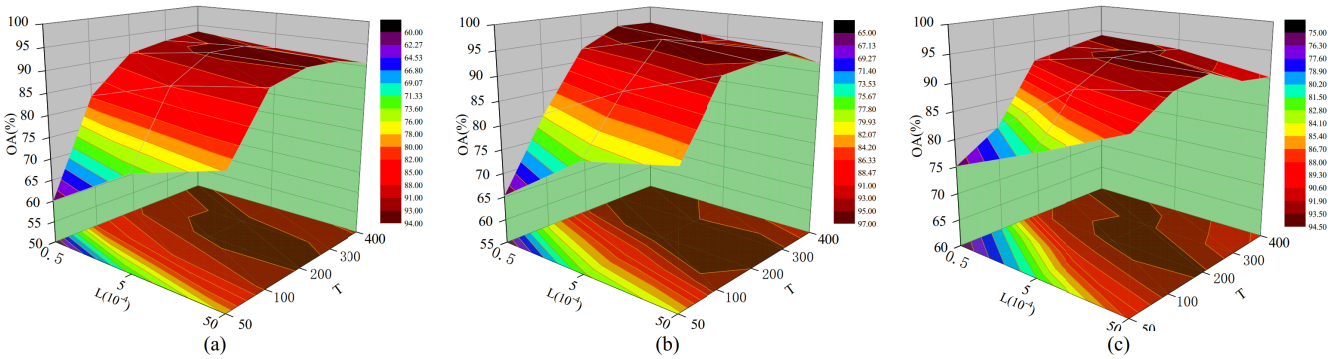


Fig. 11. Hyperparameter sensitivity analysis about the number of epochs T and learning rate L on (a) IP, (b) Salinas, and (c) UH2013.

a grid search strategy is employed, and the indicator OA is utilized to record.

The number of epochs T and learning rate L control the number of cycles of the algorithm and the step size of parameter optimization, respectively. Therefore, we conducted some experiments to access the sensitivity of two paraments varying T from 50 to 400 and L from 5×10^{-5} to 5×10^{-3}. In addition, the other parameters are fixed as in Table II. From the results in Fig. 11, we conclude that the following hold.

1) The model can train faster with a bigger L, but it is not easy to find the optimal parameters.
2) A larger T can make the model more fully trained, but it is easy to lead to over fitting. A smaller T will lead to insufficient optimization of the algorithm parameters. Therefore, it is very important to select the appropriate T.
3) T and L are closely related. The selection of one parameter affects the decision of the other. To fully optimize the parameters, while ensuring the training efficiency, it is appropriate to select the value of T and L as 200 and 5×10^{-4} from the result analysis.

The patch size P_n is closely related to the computational complexity and the amount of spatial information preserved by the patch. In this article, a different P_n impacts on HSI classification are investigated, and the results are shown in Fig. 12(a). In general, larger patches can preserve more local spatial information, while, accompanied by more noise, the classification accuracy is reduced. On the contrary, a smaller patch size will increase the number of patches, increasing the computational complexity of the algorithm. From Fig. 12(a), we note that the proposed method achieves the highest classification accuracy, while the patch size is the smallest with

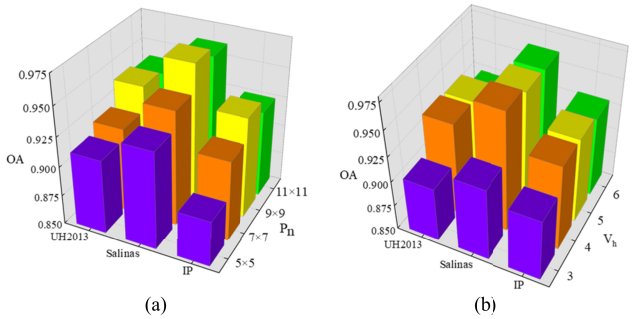


Fig. 12. Hyperparameter sensitivity analyses on three datasets. (a) Patch size P_n. (b) Number of heads V_h.

a 9×9 patch size. The SA head can extract global information from the image. After that, we analyzed the effect of the number of SA heads in S^{2}GFormer on HSI classification. From the results in Fig. 12(b), we can observe that the classification accuracy of the algorithm is the highest as $V_h = 4$. The results indicate that excessive to extract the global feature information of HSI is not beneficial to improve the classifier performance.

F. Ablation Studies

The proposed S^{2}GFormer mainly contains three modules: including pixel to region transform module, patchwise spectral embedding block, and multilayer Graphormer Encoder module. Also, each novel component plays a pivotal role in the proposed method. In this section, to see how they contribute to the overall classification performance, we conduct a set of ablation studies. Thus, S^{2}GFormer without the pixel-to-region transform module is called S^{2}GFormer-V1. In addition, S^{2}GFormer-V2 is obtained by replacing the patchwise spectral embedding block with the mean operation. Moreover,

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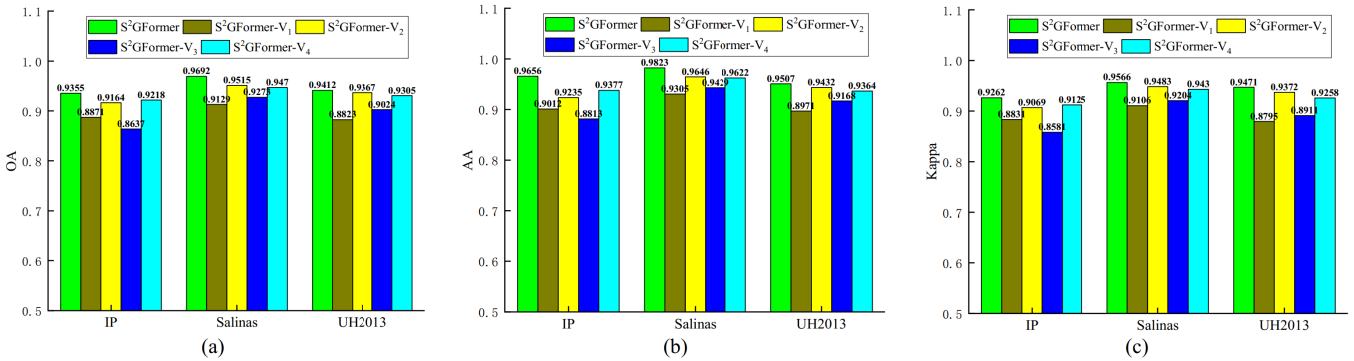


Fig. 13. Ablation experimental results on IP, Salinas, and UH2013. (a) OA. (b) AA. (c) Kappa.

the S²GFormer-V3 is obtained by replacing the Graphormer Encoder module with a conventional Transformer. Furthermore, S²GFormer-V4 is formed by removing the residual block in the Graphormer Encoder. The classification results are shown in Fig. 13. From the result, we can observe that: 1) compared with the S²GFormer, the classification accuracies of all compared methods are decreased in all indicators; 2) the index decreases to varying degrees in different simplification methods, and the magnitude varies, which means that each module designed in our method has a different impact on the classification results; and 3) all modules proposed in S²GFormer contribute to classification accuracy improving.

In addition, the role of graph residual block in Graphormer Encoder is investigated. Specifically, the residual block is removed in different Graphormer Encoders on the IP, Salinas, and UH2013 datasets. We show the results in Table VI, which indicates that graph residual block improves the classification accuracy compared with a base graph convolutional layer.

G. Comparison of Computational Efficiency

In this section, the efficiency and computational consumption of investigated methods are compared, in which the training time, testing time, and floating point operations (FLOPS) are tested. From the results in Table VII, we can conclude that the training and testing time of S²GFormer are acceptable compared with investigated networks. Due to the follow patch designed in our proposed method, the complexity of the S²GFormer is relatively tiny, and FLOPs reach a lower level. Therefore, S²GFormer reduces computational complexity while achieving optimal classification accuracy.

V. Conclusion

This article proposes a unique Graphorme with spatial–spectral features for HSI classification. The design of the follow patch mechanism can transform the pixel in HSI to locality-preserving regions, and the computational complexity of S²GFormer can be reduced. After that, a patchwise spectral embedding block is developed to extract comprehensive spectral information for patches with a newly designed neighborhood convolution. Finally, the multilayer Graphormer Encoder module, combining graph convolution, and Transformer structures into a unified Transformer, is proposed to extract the representative spatial–spectral features from the patch for HSI classification. In our method, the three modules are jointly unified together into an end-to-end network, and each component benefits the other.

In the future, we will apply more techniques, e.g., self-supervised learning and reinforcement learning, to Graphormer to improve its performance on the HSI classification task. Moreover, we will establish more integration forms of graph convolution and Transformer, and explore the implementation of graph node coding in the Transformer coding.

Acknowledgment

The authors gratefully thank the Editor and Reviewer for your outstanding comments and suggestions, which greatly helped them to improve the technical content and presentation of this article.

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